The Islamic University of Gaza
Faculty of Engineering
Department of Computer Engineering

MedLink

Web Development Project Proposal

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MedLink Team

Our team is a cohesive and well-integrated unit, combining diverse skills and perspectives to achieve a common goal. Each member contributes unique strengths, which enables effective collaboration and ensures that all aspects of the project are developed with high quality and consistency.

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Problem Description

Many people face difficulties in finding the medications they need, especially during drug shortages or emergency situations. In many cases, patients or their family members have to visit multiple pharmacies in search of a specific medication, which leads to a waste of time and effort and may negatively impact the patient's health.

This problem arises due to the lack of a centralized system that provides information about medication availability across different pharmacies, as well as the absence of an efficient way for pharmacies to share their available stock with the public in real-time.

Additionally, the communication between citizens and pharmacies is still traditional and unorganized, making it harder to access medications when needed. Therefore, there is a strong need for a digital solution that connects both parties and provides accurate and up-to-date information about drug availability.

Target Users

The system targets three main types of users:

Citizens (General Users):

These are individuals who are searching for specific medications. They can use the system to search for drugs, view the pharmacies that have them available, and post requests if the medication is not found.

Pharmacies

Pharmacies can register in the system to manage and display their available medications. They can add new drugs, update quantities, and respond to user requests, improving communication with the public.

System Administrator (Admin):

The administrator is responsible for managing the entire system. Their tasks include managing users, approving pharmacy registrations, monitoring data, and ensuring the accuracy of the information within the platform.

Main Features

The system includes a set of core features designed to facilitate access to medications and improve communication between users and pharmacies:

Medication Search:

Users can search for a specific medication by name and view a list of pharmacies where it is available, along with availability details.

Display Available Medications:

Pharmacies can display a list of available medications along with their quantities, allowing users to easily check availability.

Medication Request System:

If a medication is not available, users can post a request. Pharmacies can view these requests and respond if they have the medication.

Pharmacy Account Management:

Pharmacies can create and manage their accounts, including adding, updating, and deleting medications.

Real-Time Data Updates:

The system allows continuous updates of medication availability and quantities to ensure accurate and up-to-date information.

System Architecture Overview

The system consists of three main layers:

Frontend

- Built using HTML, CSS, and JavaScript.
- Enables citizens and pharmacies to interact with the platform, such as searching for medicines, displaying available medicines, and submitting requests.
- Provides a user-friendly and fast experience on both desktop and mobile devices.

Backend

- Developed with PHP using the Laravel framework.
- Handles all system logic, such as user registration, pharmacy management, processing requests, and responding to user requests.
- Ensures data security and different access permissions for each user type (citizen, pharmacy, admin).

Database

- Uses MySQL to store user data, medicines, requests, and stock updates.
- Supports fast and efficient search and query operations.

Data Flow
A user searches for a medicine through the frontend → the request is sent to the backend → the backend queries the database → the results are displayed to the user.

When a new medicine request is submitted, it is stored in the database and pharmacies are notified to respond.

Technology Stack

Frontend

- HTML– for structuring web pages
- CSS – for styling and designing the user interface
- JavaScript – for adding interactivity and dynamic behavior

Backend

- PHP with Laravel framework – for handling requests and managing system logic

Database

- MySQL – for storing users, pharmacies, medicines, and requests data

Expected Challenges

1. Ensuring real-time accuracy of medicine availability across pharmacies
2. Managing data consistency when multiple pharmacies update their stock simultaneously
3. Designing a user-friendly interface for different types of users
4. Handling a large number of search requests efficiently
5. Securing user data and preventing unauthorized access
6. Encouraging pharmacies to regularly update their data
7. Integrating future features such as notifications and location services
8. Testing and debugging the system to ensure stability and performance

Project Timeline

Week 1-2 25 - 3 - 2026	**Project proposal + Requirements analysis(user stories)**
Week 3	**System architecture design**
Week 4	**UI wireframes**
Week 5	**HTML pages**
Week 6	**CSS styling and responsive layout**
Week 7	**JavaScript interactivity**

MidLink
Web app